A13481W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2: SYMMETRY AND RELATIVITY

TRINITY TERM 2021

Wednesday, 16 June

Opening time: 9.30 am UK Time

You have two hours writing time to complete the paper
and up to 30 minutes technical time to upload your answer file.

Answer **two** *questions.*

Write your candidate number, paper name, paper code and a list of the questions that you have answered on the first page, and submit all of your answers as a single PDF file.

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors will be available within Inspera during the examination

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Page 2 contains physics formulae and data for this paper.
The questions start on page 3.

The following notation is used throughout the paper: capital bold letters (e.g. **U**) indicate 4-vectors; small case bold letters (e.g. **v**) indicate 3-vectors.

However the symbols **E** **B** and **A** exceptionally indicate 3-vectors, being the electric and magnetic fields and the magnetic vector potential, respectively.

The Einstein summation convention, the Minkowski metric, and other conventions adopted and used in the course are applied.

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1. (a) An electron and a positron are initially at rest at $x = 0$ and $x = +d$, respectively, in the lab frame S. At $t = 0$, a constant electric field E is switched on in the $+x$ direction. Draw the world lines of both particles on a spacetime diagram, carefully explaining your reasoning. [NB: you may ignore the direct interaction between the electron and the positron.] [4]

(b) Now consider a second frame S', moving at velocity v along the $+x$-direction of S. Using your spacetime diagram, or otherwise, describe the motion of the two particles in S'. Comment on the time-dependence of the total 3-momentum in each frame. Is total 3-momentum always conserved, and does it always transform as part of a 4-vector? [8]

(c) If instead the electric field is in the $-x$ direction, the electron and positron are accelerated towards each other until they collide and are are anihilated, resulting in the production of N photons. If $d = 20\,\mathrm{cm}$ and $E = 50\,\mathrm{kV\,cm^{-1}}$, what is the kinetic energy of the electron and positron at the time of the collision in frame S? [NB: The electron and positron have rest mass $m_{\mathrm{e}} = 0.511\,\mathrm{MeV}\,c^{-2}$.] What is the minimum value of N and the corresponding wavelength of the photons? Justify your answers. [7]

(d) If the collision is energetic enough, it can result instead in a pair of massive particles of opposite charge, for example W-bosons: $\mathrm{e}^+ + \mathrm{e}^- \rightarrow \mathrm{W}^+ + \mathrm{W}^-$. If the electric field is unchanged, how far apart would the electron and positron need to start off for this reaction to be kinematically allowed? [NB: The W-bosons have rest mass $m_{\mathrm{W}} = 80.379\,\mathrm{GeV}\,c^{-2}$.] Briefly discuss the pros and cons of large-scale linear accelerators, as opposed to circular colliders for studying such heavy particles. [6]

2. (a) Give a 4-vector argument to show that the 4-vector potential of a point charge q in an arbitrary state of motion is given by:

$$A^\mu = \frac{q}{4\pi\epsilon_0}\frac{U^\mu/c}{-R_\nu U^\nu}$$

where U^μ and R^μ are suitably chosen 4-vectors which you should define in your answer. Without detailed derivation, describe how this equation may be used to obtain an expression for the electric field of an arbitrarily moving charge. Why do we talk of a radiative and a non-radiative field? [10]

(b) A magnetic field of $B = 0.02\,\mathrm{T}$ is used to maintain short bunches of electrons in their orbits around a synchrotron of radius $r = 500\,\mathrm{m}$. Evaluate the Lorentz factor γ of the electrons and the total energy E per electron. Find the radiative energy loss per revolution. [You may use without further justification Larmor's formula

$$P_L = \frac{2}{3}\frac{q^2}{4\pi\epsilon_0}\frac{a_0^2}{c^3}.$$

for the power emitted by a charge q undergoing proper acceleration a_0.] [7]

(c) A detector located some distance away from the synchrotron can only receive radiation propagating in the x-direction. Show that the detector would receive pulses of electromagnetic radiation, and estimate the duration of those pulses. [8]

3. (a) In the lab frame S an electron is released from rest into a region with uniform and static electric field $\mathbf{E} = (E, 0, 0)$ and magnetic field $\mathbf{B} = (0, B, 0)$. Which frame S' is best-suited to work out the trajectory of the electron if (i) $|E/B| > c$ and (ii) $|E/B| < c$, and why? In each case, describe and/or sketch the trajectory of the electron in frame S' and then in the lab frame. A detailed mathematical derivation is not required, but you should carefully explain your reasoning. [7]

(b) The stress-energy tensor of the electromagentic field may be written

$$T^{\mu\nu} = \epsilon_0 c^2 \left(-F^{\mu\lambda} F_{\lambda}{}^{\nu} - \frac{1}{4} g^{\mu\nu} F_{\kappa\lambda} F^{\kappa\lambda} \right)$$

where $g^{\mu\nu}$ is the metric, and $F^{\mu\nu}$ is the electromagnetic field tensor. Find the top row of the stress-energy tensor ($T^{0\nu}$) in terms of $\mathbf{E}$ and $\mathbf{B}$ for a general field, carefully explaining each step of your derivation, and identify the physical quantities obtained. [5]

(c) A parallel-plate capacitor has its plates parallel to the xy plane and moves relative to the laboratory in the x direction at speed v. By writing down the Faraday tensor in the capacitor's rest frame, where the electric field between the plates is E, obtain the stress-energy tensor first in the rest frame, and then in the laboratory. Repeat the procedure if the capacitor has its plates parallel to the yz plane, but still moves in the x direction. Use your results to discuss how the energy stored in the capacitor is transported in each case. [13]

4. (a) What is a symmetry, and which symmetry is at the heart of Special Relativity? What is an easy way to tell if an equation is consistent with Special Relativity? [5]

(b) Write down a matrix representing a clock-wise rotation by angle θ about the origin in two dimensions, and use it to show that such rotations are orthogonal. Then consider a Lorentz boost with rapidity η in one dimension, and show that it is not orthogonal. [6]

(c) Show that the matrix

$$J_3 = \begin{bmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

gives rise to a rotation about the z axis, using the notation $R_z(\theta) = \mathrm{e}^{-\mathrm{i}\theta J_3}$. More generally, a rotation by angle θ about the axis defined by unit vector $\hat{\mathbf{n}}$ can be written in the form

$$R = \mathrm{e}^{-\mathrm{i}\theta \hat{n}^k J_k}$$

where the J_k are the generators of rotations in three dimensions. What does the commutator between these generators represent? Using this commutator, show that rotations in three dimensions form a group. [8]

(d) Without detailed derivation, but with reference to your answer to the previous part of the question, explain why Lorentz boosts in more than one dimension do not form a group. What is the smallest group which contains Lorentz transformations in all three dimensions, and by which transformations is it connected to the other divisions of the wider Lorentz group? [6]